

NARCOSCAN

Context-Aware Multimodal Narcotics Screening & Railway Threat Intelligence System

1. Executive Positioning

Core idea: Do not compete with mature commercial Raman/IMS instruments by claiming to be a better chemical identifier. Instead, build a railway-specific screening ecosystem that combines active interrogation, multimodal response analysis, environmental compensation, spatial correlation, AI risk scoring, offline operation and centralized RPF monitoring.

One-line differentiation: Existing systems primarily identify a substance at the point of measurement; NARCOSCAN creates a context-aware chemical screening network that identifies suspicious anomalies across crowded railway environments and automatically escalates them to centralized monitoring.

2. Proposed Detection Architecture

TARGET / AREA	↓	Active interrogation
Active interrogation	→	Chemical + optical + environmental response
Multimodal response	→	Temporal response fingerprint
Temporal fingerprint	→	Environmental compensation
Environmental compensation	→	Spatial consistency analysis
Spatial consistency	→	Adaptive sensor fusion
Adaptive sensor fusion	→	AI risk engine
AI risk engine	→	Normal / Unknown / High-risk
High-risk event	→	GPS + timestamp + encrypted event + alert
Online / Offline	→	Central sync / local encrypted queue

3. Why the Approach Is Different

Dimension	Existing handheld approach	NARCOSCAN proposal
Primary objective	Point/sample identification	Rapid non-intrusive railway screening
Sensing	Usually one dominant modality	Multimodal sensor fusion
Measurement	Point measurement	Temporal + spatial response analysis
Environment	Device-specific mitigation	Dynamic local environmental baseline
AI	Library matching / classification	Adaptive confidence + anomaly detection
Unknowns	Often library-dependent	Explicit unknown / inconclusive class
Network	Primarily standalone	Distributed detection network
Location	Not necessarily central	Automatic GPS tagging
Offline	Varies	Offline-first event queue and later sync

Dimension	Existing handheld approach	NARCOSCAN proposal
Operations	Instrument-centric	RPF workflow + centralized control room
Analytics	Individual result	Cross-device spatial/temporal correlation

4. The Four Main Innovation Layers

A. Active multimodal interrogation

Instead of relying only on passive concentration readings, the proposed architecture observes controlled, time-varying responses across multiple sensing modalities and converts them into a response fingerprint.

B. Context-aware environmental compensation

The system continuously models temperature, humidity, background chemical signals and sensor drift. The classifier works with contextual deviation rather than blindly interpreting raw sensor values.

C. Spatial chemical response mapping

Multiple measurements can be associated with location and scan position to identify spatially consistent anomalies rather than treating every reading as an isolated point.

D. Network-level threat intelligence

Individual handheld units become nodes in a distributed sensing network. The central system correlates time, location, confidence and environmental context to identify repeated or clustered anomalies.

5. Operational Workflow

1	Operator starts screening	Minimal interaction
2	Device establishes/updates environmental baseline	Compensates for station conditions
3	Active scan acquires multimodal response	Chemical + optical + environmental data
4	Edge AI extracts response features	Works locally, including offline
5	Fusion engine calculates confidence	Normal / unknown / high-risk
6	High-risk event is generated	GPS + timestamp + device ID
7	Event is encrypted and transmitted	Immediate online alert
8	If offline, event is queued locally	Automatic synchronization later
9	Control room receives and correlates events	Map + alert + analytics

6. What We Should NOT Claim

The project should not claim that a low-cost prototype can identify every narcotic with forensic accuracy, that a positive screening result proves possession, or that one inexpensive sensor can replace established Raman/IMS/IR systems. The defensible claim is intelligent, non-intrusive screening and escalation.

7. SIH MVP Demonstration

Hardware	Embedded controller/SBC, multimodal sensing modules, GPS, display, battery, wireless connectivity
Edge software	Signal acquisition, feature extraction, environmental compensation, ML inference, risk engine
Offline	Encrypted local event database and synchronization queue
Backend	Secure API, event ingestion, alert engine, database

Dashboard	Live device status, railway map, alerts, event history, analytics
Demo sequence	Scan → classify → risk score → GPS tag → encrypted event → control-room alert → network-offline scan → automati

8. Final Positioning for Judges

“We are not trying to reinvent Raman or IMS. We are reinventing the operational screening layer around chemical detection for crowded railway environments—using multimodal response fingerprints, environmental context, spatial correlation, edge AI and a secure offline-capable RPF threat-intelligence network.”